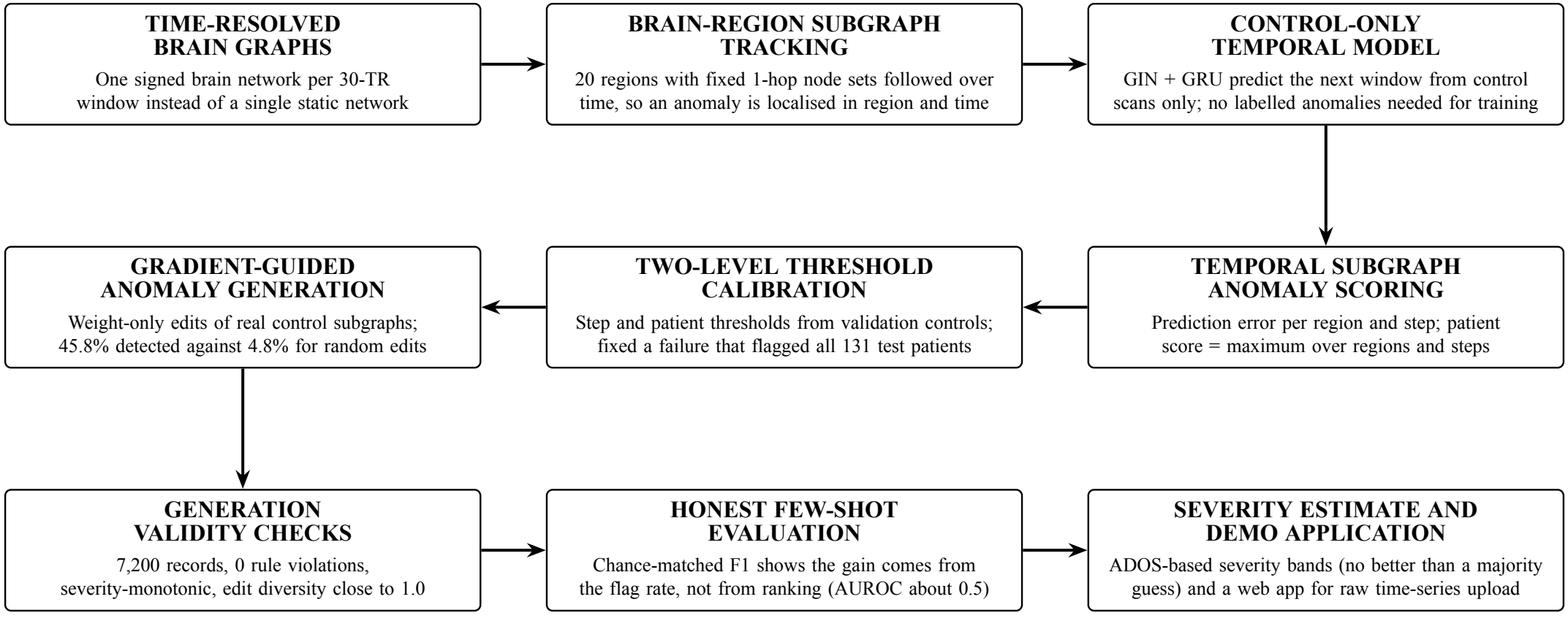
TIME-RESOLVED BRAIN GRAPHS
One signed brain network per 30-TR window instead of a single static network
BRAIN-REGION SUBGRAPH TRACKING
20 regions with fixed 1-hop node sets followed over time, so an anomaly is localised in region and time
CONTROL-ONLY TEMPORAL MODEL
GIN + GRU predict the next window from control scans only; no labelled anomalies needed for training
TEMPORAL SUBGRAPH ANOMALY SCORING
Prediction error per region and step; patient score = maximum over regions and steps
TWO-LEVEL THRESHOLD CALIBRATION
Step and patient thresholds from validation controls; fixed a failure that flagged all 131 test patients
GRADIENT-GUIDED ANOMALY GENERATION
Weight-only edits of real control subgraphs; 45.8% detected against 4.8% for random edits
GENERATION VALIDITY CHECKS
7,200 records, 0 rule violations, severity-monotonic, edit diversity close to 1.0
HONEST FEW-SHOT EVALUATION
Chance-matched F1 shows the gain comes from the flag rate, not from ranking (AUROC about 0.5)
SEVERITY ESTIMATE AND DEMO APPLICATION
ADOS-based severity bands (no better than a majority guess) and a web app for raw time-series upload